Technology Stack

Date	7 July 2026
Team ID	SWTID-2026-2018
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Technology Stack

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript	Used to create a responsive and user-friendly interface for entering applicant details, validating inputs, and displaying eligibility prediction results.
2	Backend / Server-Side	Python, Flask	Flask is used to develop the web application, handle routing, process user inputs, load the trained machine learning model, and return prediction results.
3	Data Storage / Machine Learning	CSV Dataset, Pandas DataFrames, Joblib (.pkl Files)	The CSV dataset is used for training the model, while Joblib stores the trained model, scaler, encoder, and feature columns for real-time predictions.
4	Cloud / Hosting / Deployment	Render, Gunicorn	Render is used to deploy the Flask application on the cloud, while

			Gunicorn serves as the production WSGI server for handling requests efficiently.
5	Version Control & Collaboration	Git, GitHub	Used for source code management, version control, collaboration among team members, and deployment integration.
6	Machine Learning Libraries & Development Tools	Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, OpenPyXL, Joblib	Used for data preprocessing, feature engineering, model training, evaluation, visualization, model serialization, and Excel dataset support.